

SKILLS

- **Security Engineering:** threat modeling, secure code review, vulnerability assessment, OAuth/OIDC, IAM, Zero Trust
- **AI Agents:** Python, LLM orchestration, RAG, MCP, tool use, agent evaluation, guardrails, human approval workflows
- **Software Engineering:** TypeScript, Go, SQL, React, Node.js, REST APIs, distributed systems, session management
- **Cloud & Infrastructure:** AWS, GCP, OpenShift, Kubernetes, Docker, Terraform, CI/CD, air-gapped deployment
- **Data & Operations:** Kafka, Elasticsearch, Redis, OpenTelemetry, audit trails, SLOs, alerting, incident response

EXPERIENCE

- **Royal Bank of Canada (RBC)** Toronto, Canada
Staff Software Developer & Enterprise Architect Nov 2022 – Present
 - **Secure Agentic Systems:** Building ASB Assist, a Python-orchestrated LLM + RAG system that converts natural-language requirements into compliant architecture diagrams; implementing scoped tool access, security-pattern injection, grounding checks, evaluations, guardrails, and audit trails.
 - **Agent Security Controls:** Operationalized enterprise AI governance spanning IAM, model risk, Responsible AI, data residency, regulatory controls, and auditable deployment across AWS, GCP, and air-gapped OpenShift environments.
 - **Production Platform Engineering:** Designed and shipped a patent-pending TypeScript/React platform that maps 1,000+ architecture patterns to real-time service selections and automated certification; authored 99.6% of production code and all core recommendation logic.
 - **High-Trust Infrastructure:** Architected an OpenTelemetry platform using Vector, Fluent Bit, Kafka, and Logstash that ingests 150 TB/day across 10,000+ systems, with resilient pipelines, schema contracts, access controls, and vendor-independent operations.
 - **Hands-On Technical Leadership:** Lead 5 engineers through system design, implementation, security review, testing, ADRs, and operational handoff while aligning product, security, risk, and infrastructure stakeholders.
- **HealthCard (Acquired Startup)** Remote
Blockchain Architect & First Full Stack Engineer Aug 2021 – Jul 2022
 - **Secure Full-Stack Delivery:** Built the product from scratch using React/TypeScript, Node.js, Hyperledger Fabric, and AWS EKS, with authentication boundaries, tamper-evident verification, HIPAA-aligned data handling, and complete audit logging.
 - **Security Automation:** Implemented fraud-resistant insurance validation and transformer-based document classification behind monitored APIs, reducing manual review while preserving traceability for sensitive workflows.
- **Canada Revenue Agency (CRA)** Ottawa, Canada (Remote)
Software Developer Co-op Apr 2022 – Nov 2022
 - **Secure Platform Modernization:** Modernized a federal tax platform from Apache Struts to React/TypeScript and contributed to APIs serving millions of Canadians, with WCAG compliance and automated SonarQube quality gates.

PROJECTS

- **Autonomous Security Agent Workbench:** Built a Python prototype that orchestrates specialized agents for alert triage, repository-aware secure code review, threat modeling, and vulnerability assessment; isolated tools behind typed policies, session-scoped credentials, human approval gates, and immutable audit logs.
- **Agent Security Evaluation Harness:** Created adversarial task suites for prompt injection, tool misuse, privilege escalation, sensitive-data leakage, and unsafe actions; tracked attack success, authorization violations, groundedness, latency, and regression risk with end-to-end traces.

EDUCATION

- **Dalhousie University** Halifax, Canada
MASc Applied Computer Science, GPA: 3.9/4.0 | TA: Algorithms, Advanced Cloud Computing Jan 2021 – Aug 2022
- **University of Toronto** Toronto, Canada
Enterprise Architecture Certification (3-semester program) Completed 2024
- **Gujarat Technological University** Ahmedabad, India
B.E. Computer Engineering, GPA: 3.92/4.0 Aug 2016 – Aug 2020